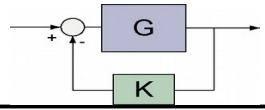


ME 2801

Introduction to Control Systems

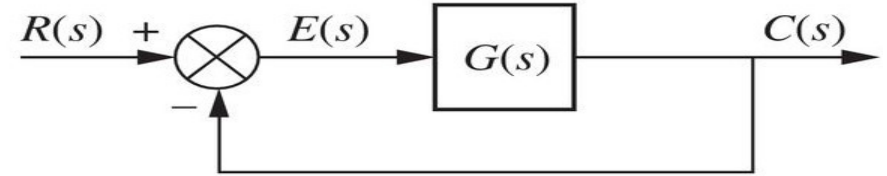
W5: Steady State Error
and System Type

Steady-State Error



So what? Closed-loop system response specifications

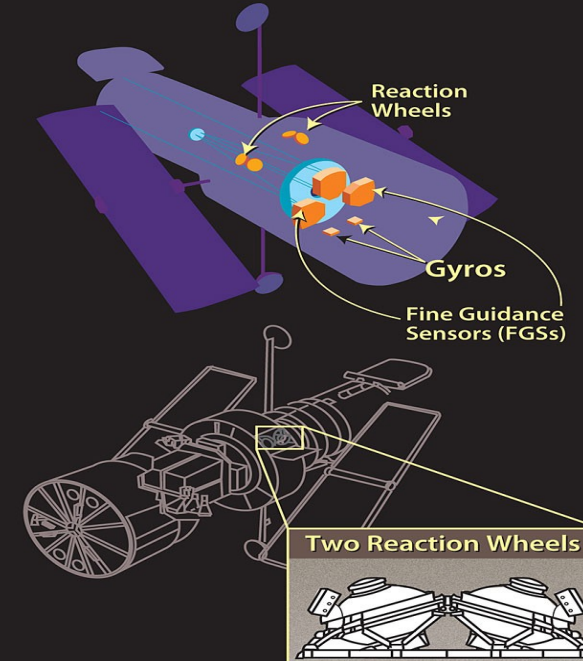
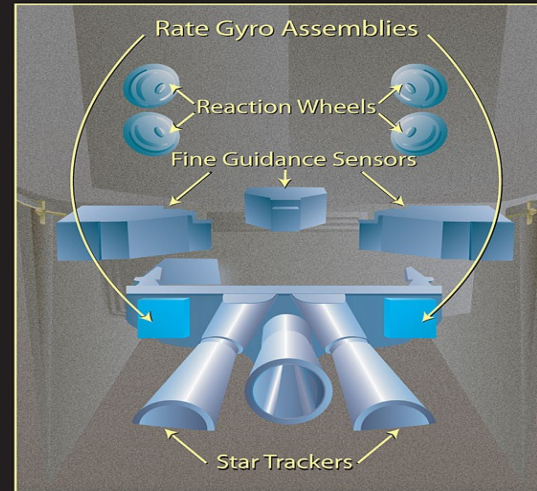
- Transient response
- Stability
- Steady-state errors



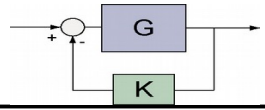
Input Type

1. Step (position)
2. Ramp (velocity)
3. Parabola (acceleration)

Hubble's Pointing Control System



Steady-State Error

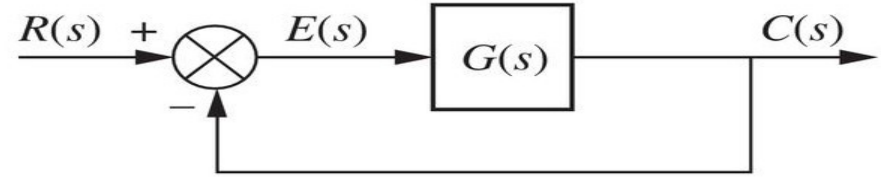


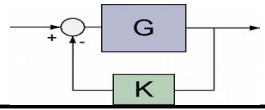
So what? Closed-loop system response specifications

- Transient response
- Stability
- Steady-state errors

Input Type

1. Step (position)
2. Ramp (velocity)
3. Parabola (acceleration)

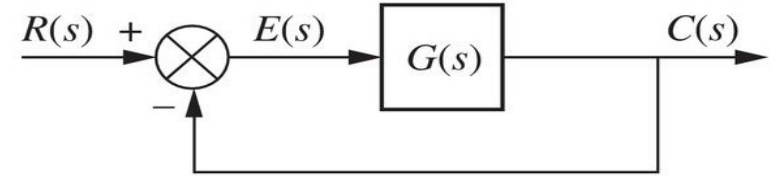




Steady-state error as performance specification for feedback system.

- Given a unity feedback system, what is the steady-state response?

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$



- For the following test inputs

- Step (position)

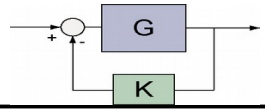
$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}$$

- Ramp (velocity)

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}$$

- Parabola (acceleration)

$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2G(s)}$$



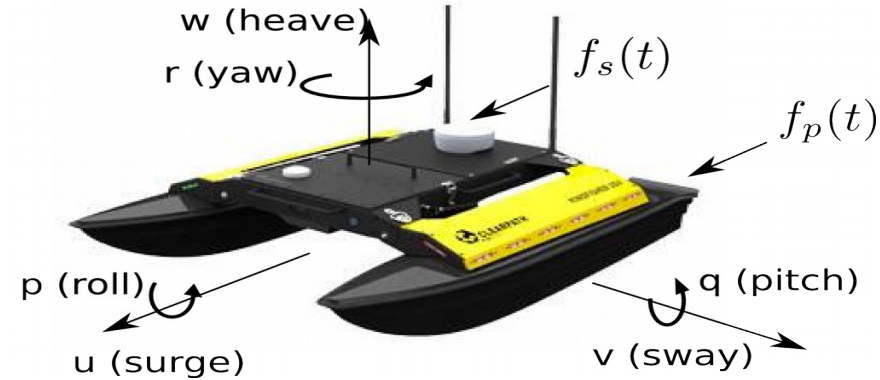
Consider steady-state error for..

- Surge proportional control
 - Step and Ramp input

$$G_u(s) = \frac{U(s)}{T(s)} = \frac{1}{Ms + D}$$

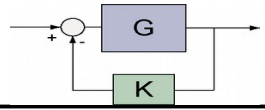
- Yaw proportional control
 - Step and Ramp input

$$G_r(s) = \frac{R(s)}{\tau(s)} = \frac{1}{s(Is + d)}$$



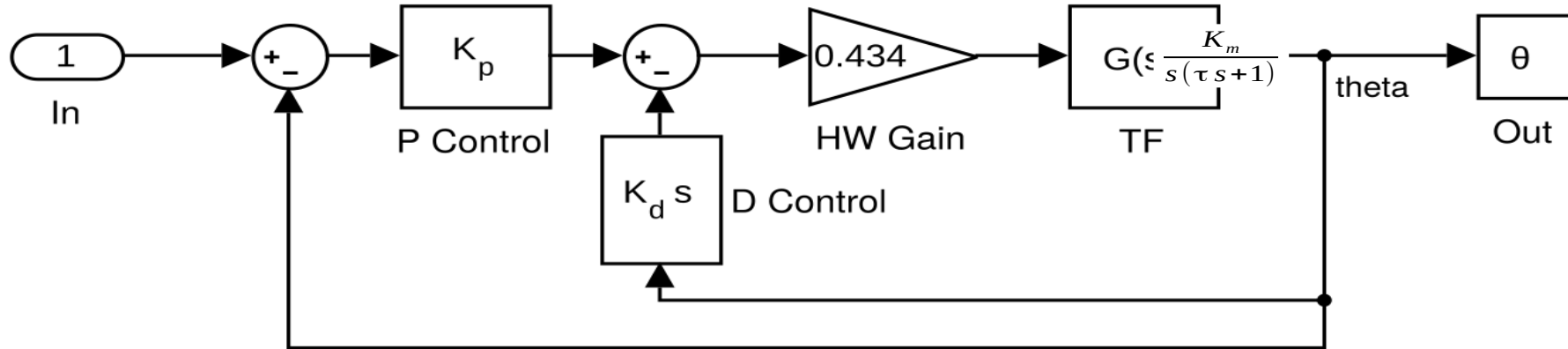
	Surge Control	Yaw Control
Step Input		
Ramp Input		

Example

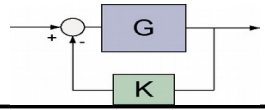


Given ECP **open-loop position** response,
what is the **closed-loop** steady state error for...

- Step, position command
- Ramp, velocity command
- Parabola, acceleration command

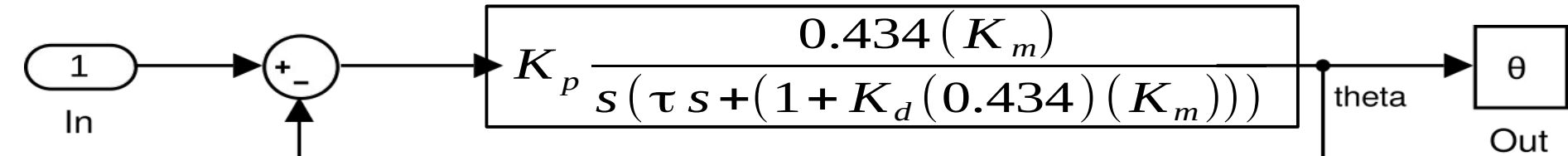


Example



Given ECP **open-loop position** response, what is the **closed-loop** steady state error for...

- Step, position command
- Ramp, velocity command
- Parabola, acceleration command

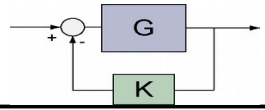


$$e(\infty) = e_{\text{step}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s)}{1 + G(s)} = \frac{1}{1 + \lim_{s \rightarrow 0} G(s)}$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^2)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s + sG(s)} = \frac{1}{\lim_{s \rightarrow 0} sG(s)}$$

$$e(\infty) = e_{\text{parabola}}(\infty) = \lim_{s \rightarrow 0} \frac{s(1/s^3)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2G(s)} = \frac{1}{\lim_{s \rightarrow 0} s^2G(s)}$$

Example



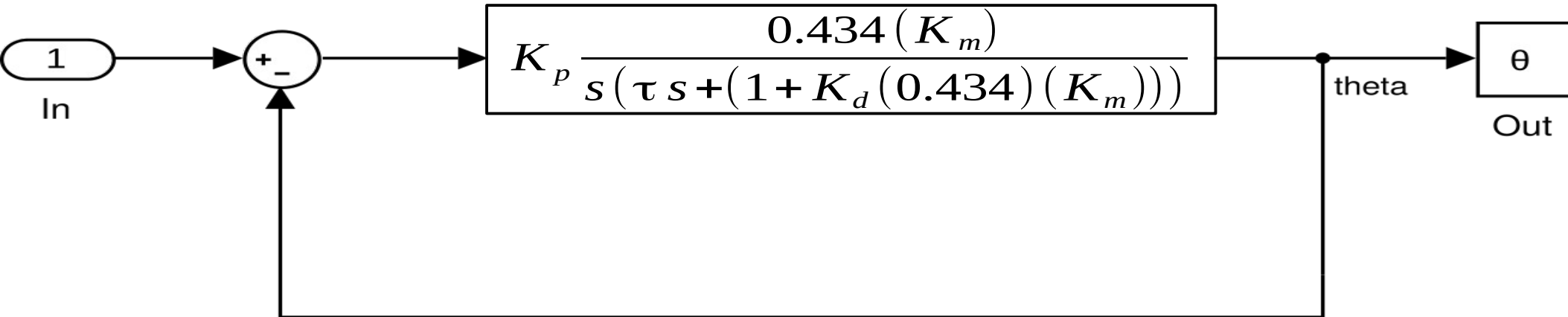
Given ECP **open-loop position** response,
what is the **closed-loop** steady state error for...

- Step, position command
- Ramp, velocity command
- Parabola, acceleration command

$$e_{step}(\infty) = 0$$

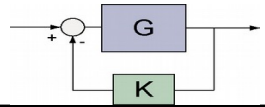
$$e_{ramp}(\infty) = \frac{1 + K_d(0.434)(K_m)}{K_p(0.434)(K_m)}$$

$$e_{parabola}(\infty) = \infty$$





System Type: Definition



System Type

- *degree of the polynomial for which the **steady-state error** is a nonzero finite constant*
- *equals the number of “pure integrators” in the open-loop transfer function.*
- We can analyze the open-loop dynamics
predicting the closed-loop tracking performance!

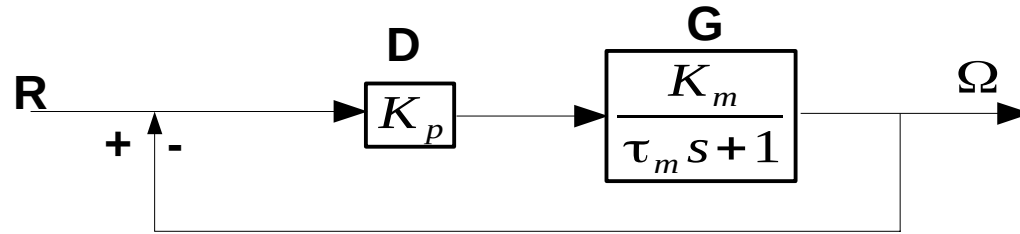
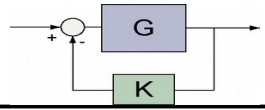
$$E = \left(\frac{1}{1 + DG} \right) R$$

So What?

- Since $D(s)$ is our controller,
we can design the controller to change the system type.
- System type can tell us, at a high level, what type of control to accomplish our goals.

“Sensitivity” TF

Example: Speed P Control



Find steady-state error $E = \frac{1}{1 + DG} R$

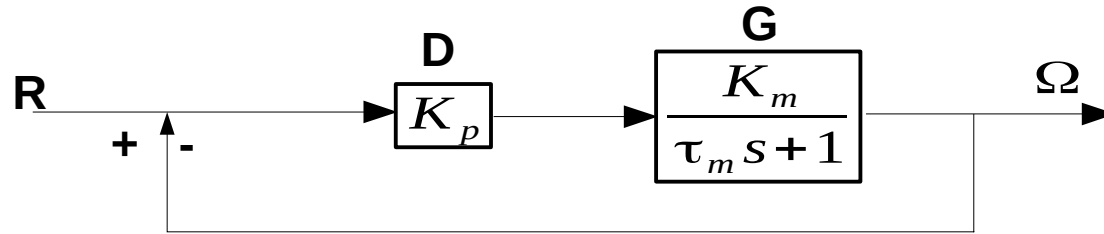
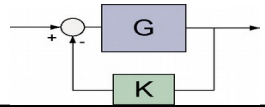
Final-value property

$$e_{ss} = \lim_{t \rightarrow \infty} e_{ss}(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \left(\frac{s}{1 + DG} \right) R$$

Now examine some test inputs: **step**, ramp, parabola

- Polynomial degree: step=0, ramp=1, parabola=2

Example: Speed P Control

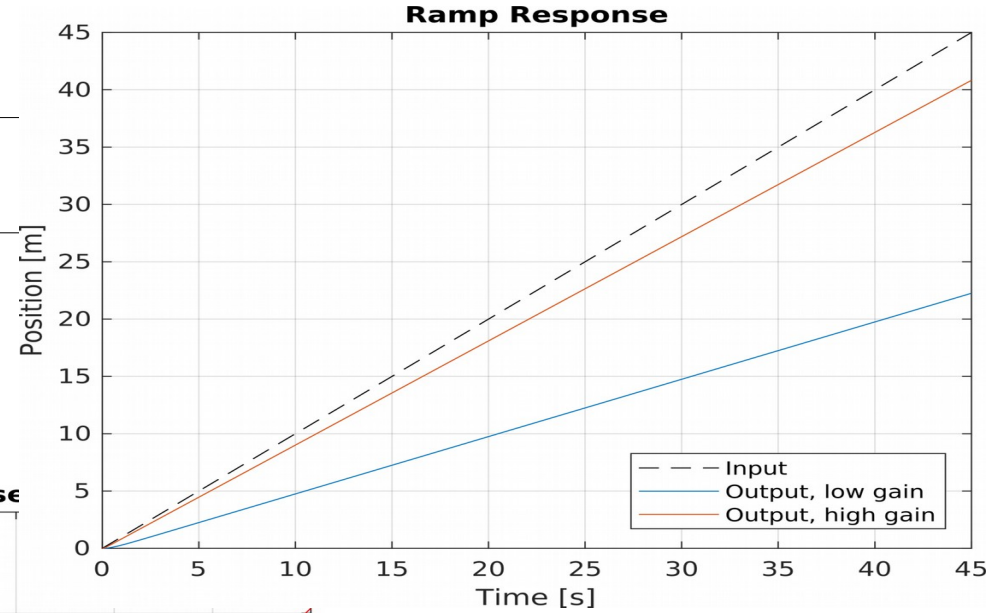
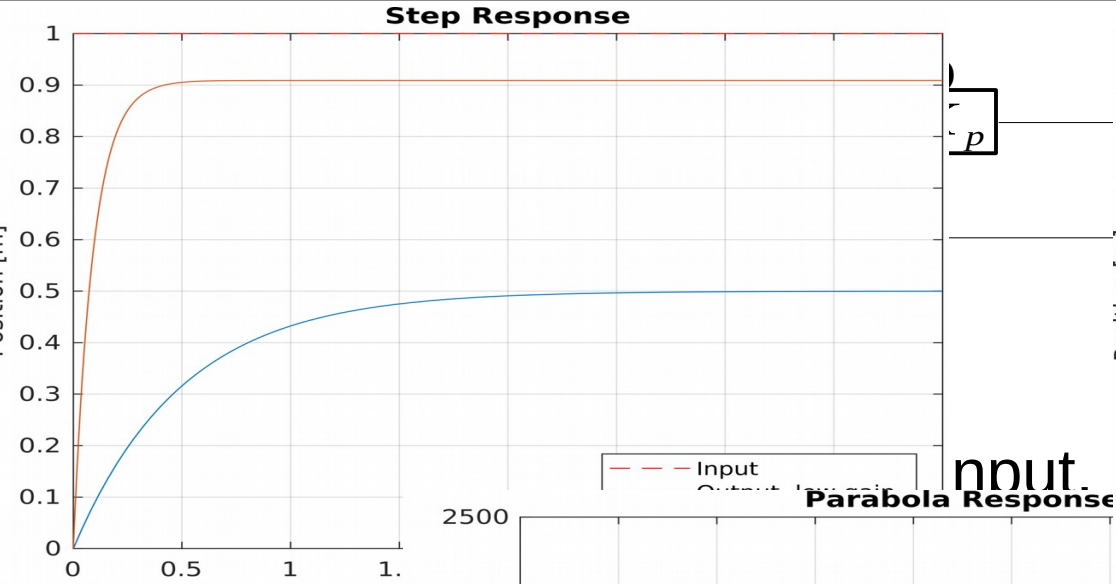
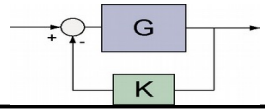


Proportional Control:

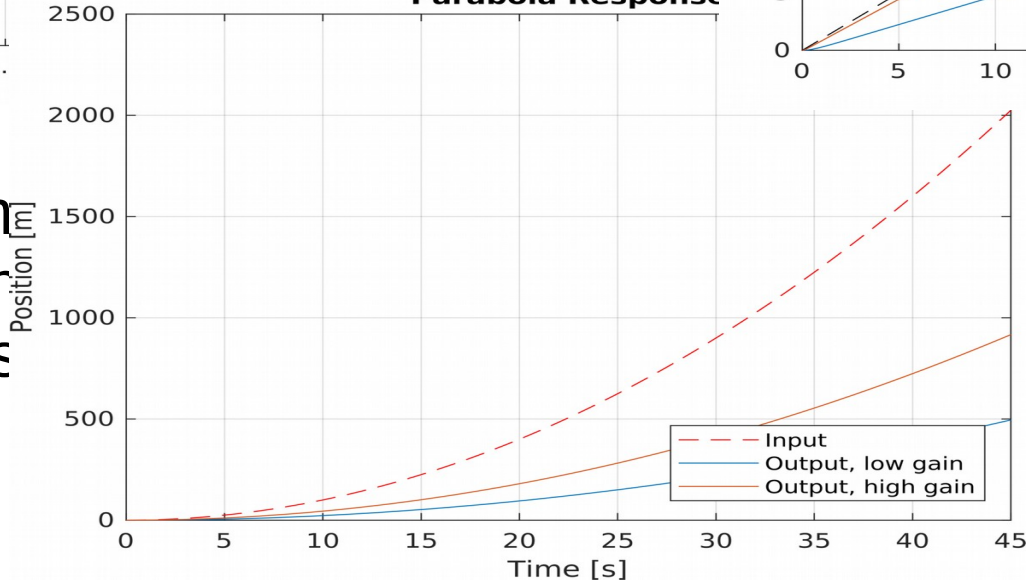
For speed output and step input,
We will always have constant,
non-zero steady-state error!

The system ($DG = L$) is type 0, meaning...
a step input (order=0) will result in constant
steady-state error.

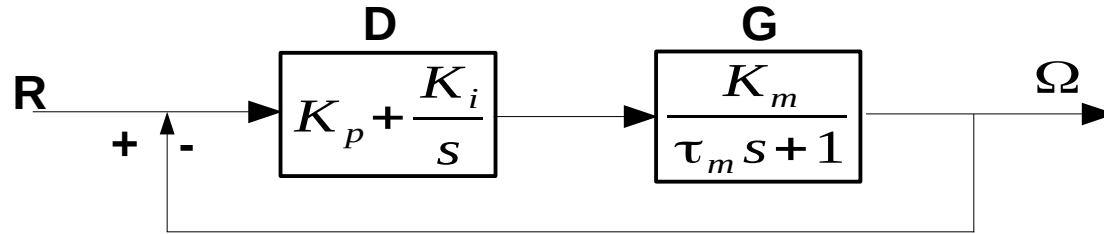
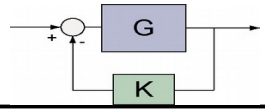
Example: Speed P Control



The system
a step in
steady-s



Example: Speed PI Control



So let's change the system type
and get rid of the steady-state error

- Add an integrator...

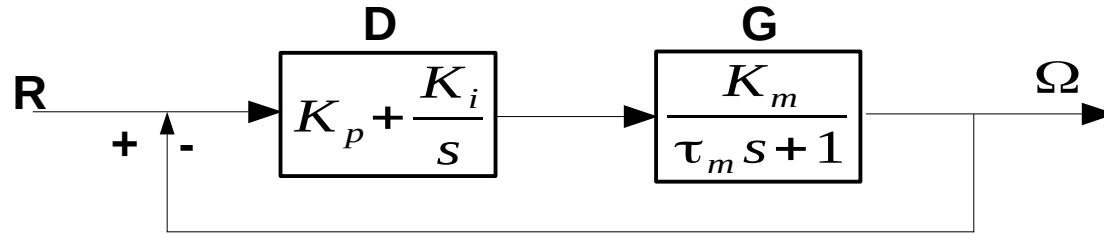
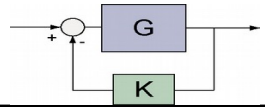
$$DG = L = \frac{K_m (K_p s + K_i)}{s (\tau_m s + 1)}$$

Again, examine some test inputs:

step, ramp, parabola

- Polynomial degree: step=0, ramp=1, parabola=2

Example: Speed PI Control



Proportional - Integral Control:

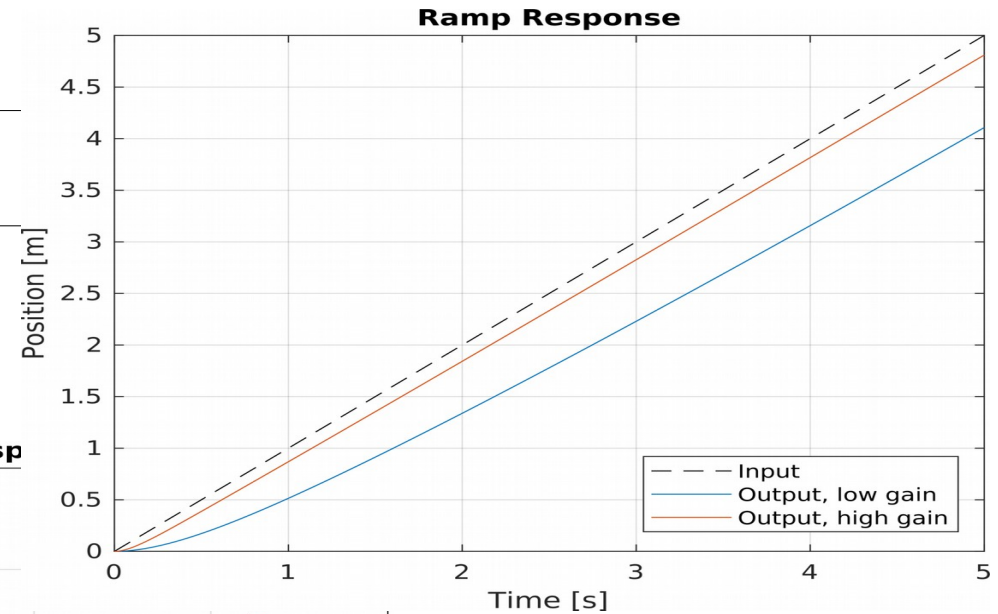
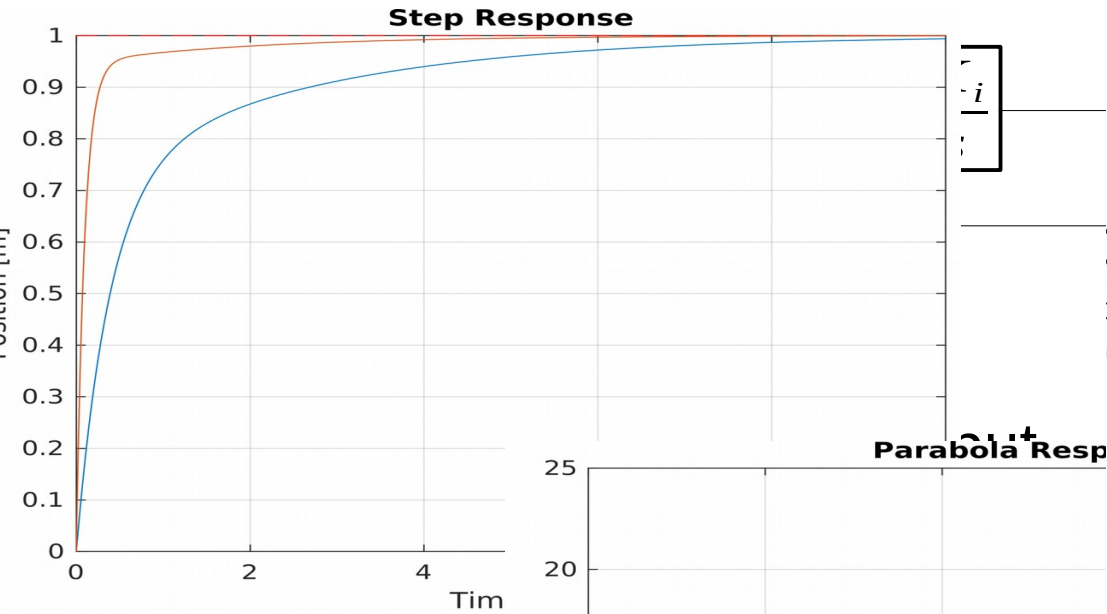
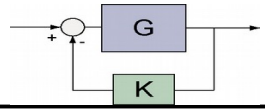
For speed output and step input,

We will always have zero steady-state error

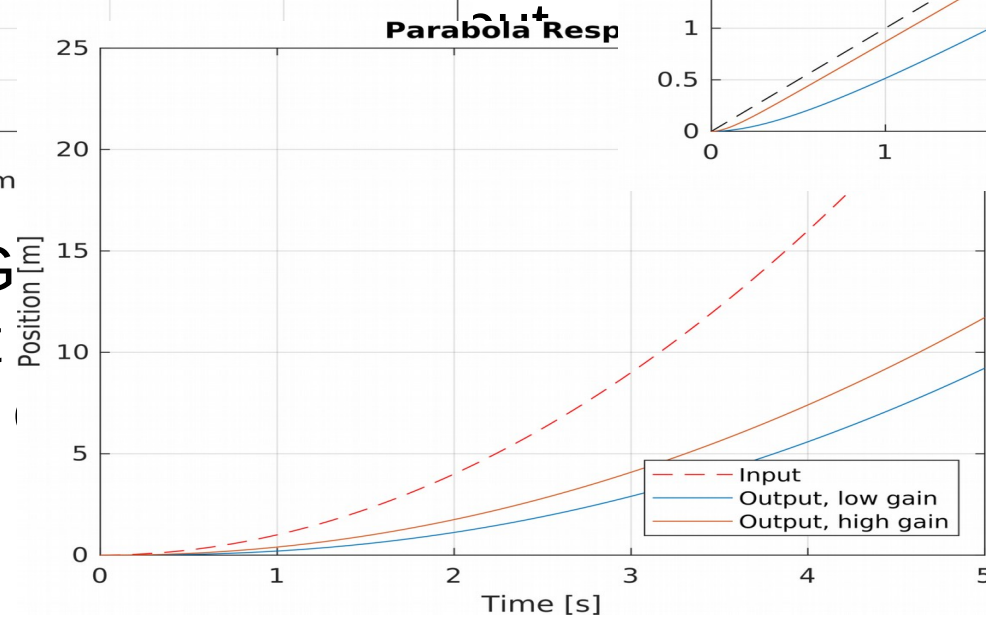
The system ($DG = L$) is type 1, meaning...

a ramp input (order=1) will result in constant steady-state error

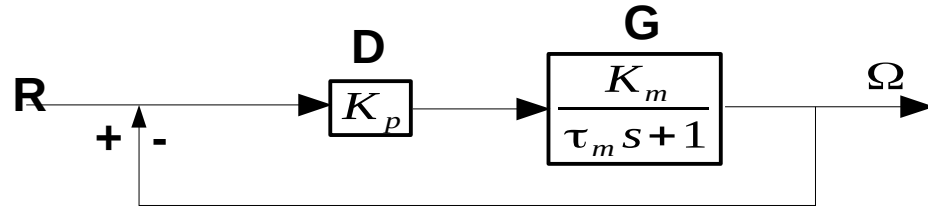
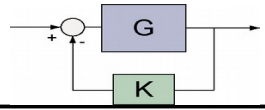
Example: Speed PI Control



The system (DG)
a ramp input
steady-state

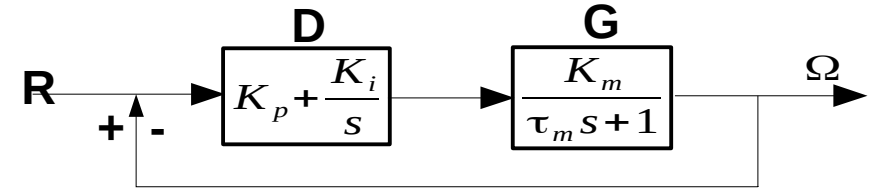
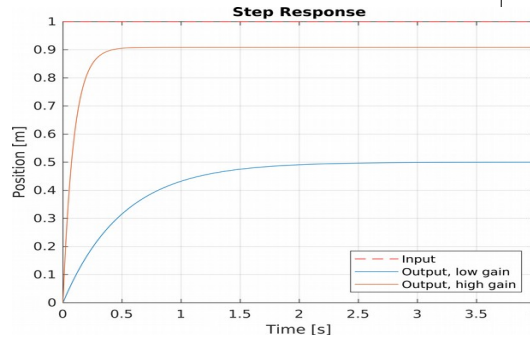


Speed Control Summary



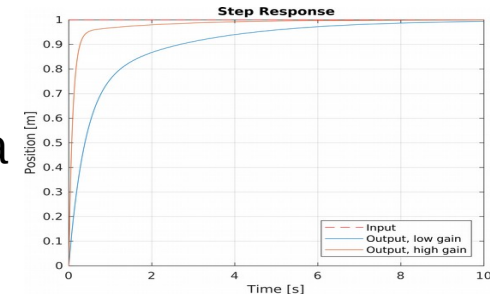
P Control

- Open-Loop System is Type 0
 - No pure integrators.
- Steady-State Error
 - Constant for step
 - Infinite for ramp
 - Infinite for parabola

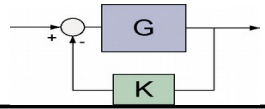


PI Control

- Open-Loop System is Type 1
 - One pure integrators.
- Steady-State Error
 - Zero for step
 - Constant for ramp
 - Infinite for parabola



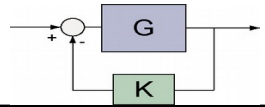
Adding an integrator changes the dynamics!



Summary of system type, inputs and constants

TABLE 7.2 Relationships between input, system type, static error constants, and steady-state errors

Input	Steady-state error formula	Type 0		Type 1		Type 2	
		Static error constant	Error	Static error constant	Error	Static error constant	Error
Step, $u(t)$	$\frac{1}{1 + K_p}$	$K_p = \text{Constant}$	$\frac{1}{1 + K_p}$	$K_p = \infty$	0	$K_p = \infty$	0
Ramp, $tu(t)$	$\frac{1}{K_v}$	$K_v = 0$	∞	$K_v = \text{Constant}$	$\frac{1}{K_v}$	$K_v = \infty$	0
Parabola, $\frac{1}{2}t^2u(t)$	$\frac{1}{K_a}$	$K_a = 0$	∞	$K_a = 0$	∞	$K_a = \text{Constant}$	$\frac{1}{K_a}$



System Type

- *degree of the polynomial for which the **steady-state error** is a nonzero finite constant*
- *equals the number of “pure integrators” in the open-loop transfer function.*

Design Takeaway

- If you want zero steady-state error,
then you need at least one pole at zero (pure integrator)
in the open-loop TF: $L = DG$